

First-in-Human, Proof-of-Mechanism Phase 1 Study of the Oral MRGPRX2 Antagonist SEP-631 Utilizing Short-Wave Infrared Imaging to Assess Response to an Icatibant Skin Challenge

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Background

- Mas-related G-protein-coupled receptor member X2 (MRGPRX2), a G-protein-coupled receptor that is predominantly expressed on human mast cells, mediates immunoglobulin E (IgE)-independent degranulation and has been implicated in several conditions, including chronic spontaneous urticaria (CSU)¹
- MRGPRX2 is activated by a host of endogenous agonists leading to IgE-independent mast cell degranulation
- Mast cell degranulation leads to the release of mediators that can cause pain, itch, inflammation, and edema
- Targeting inhibition of MRGPRX2 has the potential to disrupt feedback loops that drive these effects

MRGPRX2

Neuropeptides

- Substance P
- PACAP
- Cortistatin-14
- VIP
- Hemokinin-1
- PAMP-12

Immune ligands

- LL-37
- Defensins
- Major Basic Protein
- Eosinophil Peroxidase

Mast cell

Mast cell degranulation

Mast cell mediators

- Histamine
- Tryptase
- Chymase
- Prostaglandins
- Leukotrienes
- Cytokines
- Chemokines
- PAF

Immune cells

Capillaries

Inflammation and edema

Pain and itch

- CSU is an inflammatory skin condition marked by unpredictable exacerbations of itchy wheals (hives), angioedema, or both, that last for >6 weeks and can significantly impair daily activities and quality of life²
- There remains a significant unmet need for safe and effective CSU therapies that have the convenience and improved adherence of a once-daily oral tablet

SEP-631: Oral Small-Molecule MRGPRX2 Negative Allosteric Modulator

SEP-631 Preclinical Data Support Potential Best-in-Class Profile³

- Subnanomolar binding affinity and slow off-rate kinetics
- Insurmountable negative allosteric modulator mechanism that closes off the agonist binding pocket
 - Inability to activate receptor despite the presence of high doses of endogenous agonists
- Potent inhibition of MRGPRX2 activation in knock-in animals and primary human mast cells
- Generally well tolerated in pharmacology and Good Laboratory Practice toxicology studies
- Excellent oral pharmacokinetics (PK) across species
 - Projects to once-daily human dosing
- Excellent pharmaceutical properties
 - Low drug-drug interaction risk based on in vitro profiling
 - Low food effect risk based on preclinical studies
- Convenient tablet formulation developed

Methods

- This Phase 1, single-center, first-in-human, double-blind, placebo-controlled study evaluated the safety, tolerability, PK, and pharmacodynamics (PD) after single and multiple ascending oral doses of SEP-631 administered once daily in healthy adult volunteers (ClinicalTrials.gov Identifier: NCT07069036; **Figure 1**)

Figure 1. Study design

Part A* Single ascending dose, n=8/cohort. Randomization 6:2 (SEP-631:placebo). Doses: 10 mg, 30 mg, 90 mg, 200 mg, 300 mg, 400 mg. Key endpoints: Safety/tolerability, PK.

Part B Multiple ascending dose, n=16/cohort. Randomization 12:4 (SEP-631:placebo). Doses: 10 mg, 30 mg, 90 mg, 200 mg. Key endpoints: Safety/tolerability, PK, Skin challenge PD (Figures 2).

Part C Food effect, n=8. Randomized open-label crossover. 5-day washout. Doses: 200 mg Fasted, 200 mg Fed. Key endpoints: Safety/tolerability, PK (fasted vs fed).

*Serial dosing was used in single ascending dose cohorts. Decisions for dose escalation and progression from single to multiple doses and food-effect dosing were made by a Safety Review Committee. PO, oral administration.

- This Phase 1 study was conducted in 3 parts in healthy adult volunteers:
 - Part A: randomized, double-blind, placebo-controlled, single ascending dose (SAD) study
 - Part B: randomized, double-blind, placebo-controlled, multiple ascending dose (MAD) study
 - Part C: randomized, open-label, 2-period, crossover food-effect study
 - Single oral dose of SEP-631 was administered under fasted and fed (high-fat, high-calorie) conditions per US Food and Drug Administration guidance
- Throughout all parts of the study, participants were monitored for treatment-emergent adverse events (TEAEs)
- Blood samples were collected for assessment of PK parameters
- In Part B, PD were assessed using intradermal injection as a skin challenge test at pre-dose baseline (Day -1) and steady state following SEP-631 or placebo treatment (Day 9)
 - Skin challenge agents were sterile saline (injection control), histamine 10 µg/mL (skin wheal positive control), and icatibant 10 µg/mL and 100 µg/mL (selective MRGPRX2 agonist at low and high concentrations, respectively; **Figure 2A**).⁴ Icatibant injections were performed in duplicate to provide replicate measurements
 - Skin wheals were imaged using the AllergyScope™ (**Figure 2B**) precision digital allergy skin test wheal detector that utilizes a high-performance semiconductor alloy composed of indium, gallium, and arsenic (InGaAs) that is sensitive to light in the short-wave infrared (SWIR) spectrum from about 900 to 1700 nm. The SWIR photosensor has been shown to be superior to other imaging detectors of skin wheals (e.g., visible range, near-infrared [NIR], or thermal sensors; **Figure 2C**)⁵
 - Skin wheal image capture using the AllergyScope™⁴ detector enabled systematic precision central laboratory adjudication of wheal size by agreement of measurements performed by 2 independent and blinded adjudicators at Johns Hopkins University

Figure 2. Intradermal injections and AllergyScope™ detector for skin color-agnostic quantification of icatibant-induced wheal dimensions using SWIR

A. Pattern of intradermal forearm injections. B. Developed and patented by FLARE Diagnostics (Middle River, MD, USA), the AllergyScope™ detector produces high-resolution images of skin test reactions that allow accurate wheal measurements and blinded interpretation. Digital intradermal wheal images produced by the AllergyScope™ detector were transmitted for analysis by adjudicators blinded to treatment assignment. C. SWIR is superior to visible range, NIR, or thermal sensors for imaging skin wheals.

Results

Healthy Adult Participants

- Overall, 120 participants were dosed in the study (Part A: n=48; Part B: n=64; Part C: n=8)
- Baseline characteristics were generally comparable across treatment groups in all parts
 - Mean (standard deviation) age was 29.3 (6.92), 29.6 (6.47), and 27.3 (5.99) years in Parts A, B, and C, respectively
 - The proportion of males was 47.9%, 65.6%, and 87.5% in Parts A, B, and C, respectively

SEP-631 Adverse Event (AE) Profile Was Comparable With Placebo

- In Part A (SAD), the rate of TEAEs with single doses of SEP-631 was comparable with placebo. No severe or serious events were reported (**Table 1**)
- Two AEs of transaminase elevation with SEP-631 were mild (<1.5× upper limit of normal [ULN]), were not dose related, and occurred at rates comparable with placebo

Table 1. Part A (SAD) summary of TEAEs

	Pooled placebo (n=12)	10 mg (n=6)	30 mg (n=6)	90 mg (n=6)	200 mg (n=6)	300 mg (n=6)	400 mg (n=6)	Pooled SEP-631 (n=36)
Any TEAEs, n (%)	5 (41.7)	2 (33.3)	4 (66.7)	0	4 (66.7)	3 (50.0)	1 (16.7)	14 (38.9)
Mild	5 (41.7)	2 (33.3)	3 (50.0)	0	4 (66.7)	2 (33.3)	1 (16.7)	12 (33.3)
Moderate	0	0	1 (16.7)	0	0	1 (16.7)	0	2 (5.6)
Severe	0	0	0	0	0	0	0	0
Serious TEAEs, n (%)	0	0	0	0	0	0	0	0
TEAEs in ≥1 participant, n (%)								
Headache	1 (8.3)	0	2 (33.3)	0	1 (16.7)	1 (16.7)	0	4 (11.1)
Transaminases increased	1 (8.3)	1 (16.7)	1 (16.7)	0	0	0	0	2 (5.6)
Dysmenorrhea	0	0	1 (16.7)	0	0	1 (16.7)	0	2 (5.6)

- In Part B (MAD), the rate of TEAEs with multiple doses of SEP-631 was comparable with placebo, with no severe or serious events reported (**Table 2**)
- One mild transaminase elevation (<1.5× ULN) was observed with SEP-631, and 1 was observed with placebo

Table 2. Part B (MAD) summary of TEAEs

	Pooled placebo (n=16)	10 mg (n=12)	30 mg (n=12)	90 mg (n=12)	200 mg (n=12)	Pooled SEP-631 (n=48)
Any TEAEs, n (%)	10 (62.5)	3 (25.0)	6 (50.0)	5 (41.7)	4 (33.3)	18 (37.5)
Mild	10 (62.5)	3 (25.0)	6 (50.0)	4 (33.3)	4 (33.3)	17 (35.4)
Moderate	0	0	0	1 (8.3)	0	1 (2.1)
Severe	0	0	0	0	0	0
Serious TEAEs, n (%)	0	0	0	0	0	0
TEAEs in ≥1 participant, n (%)						
Headache	4 (25.0)	0	2 (16.7)	2 (16.7)	1 (8.3)	5 (10.4)
Dizziness	1 (6.3)	0	0	2 (16.7)	1 (8.3)	3 (6.3)
Abdominal pain	0	2 (16.7)	0	1 (8.3)	0	3 (6.3)
Nausea	1 (6.3)	0	0	1 (8.3)	1 (8.3)	2 (4.2)
Fatigue	1 (6.3)	0	0	1 (8.3)	0	1 (2.1)
Dermatitis contact	1 (6.3)	0	1 (8.3)	0	0	1 (2.1)
Alanine aminotransferase or Transaminases increased	1 (6.3)	0	0	0	1 (8.3)	1 (2.1)
Dysmenorrhea	1 (6.3)	0	1 (8.3)	0	0	1 (2.1)

Pharmacokinetics: Dose-Proportional, Once-Daily Dosing Without Food Effects

- Mean plasma concentration-time profiles following multiple doses of once-daily SEP-631 (Part B, MAD) shown in **Figure 3**
- Approximately dose-proportional increases in exposure were observed across the evaluated dose range
- The steady-state half-life of SEP-631 is approximately 24 hours, supporting once-daily dosing
- In a dedicated food-effect study, SEP-631 administered with a high-fat, high-calorie meal showed similar exposure to fasted conditions (area under the curve [AUC] and maximum plasma concentration [C_{max}] ratios of approximately 90%), supporting dosing without food restrictions

Figure 3. Mean SEP-631 plasma concentration-time profiles for Part B (MAD) cohorts on Day 10

Proof of Mechanism: Potent Inhibition of Icatibant-Induced Skin Wheals

- At baseline (Day -1), high-precision wheal measurements were obtained with the AllergyScope™ detector
- Independent wheal measurements obtained by the AllergyScope™ detector yielded between-reader precision of 6% CV relative to the histamine positive control wheal size, indicating high reproducibility not possible with traditional manual skin test reading (**Figures 4, 5**)
- In the skin challenge at Day 9, SEP-631 showed complete inhibition of icatibant 10 µg/mL-induced wheals at doses as low as 10 mg QD (**Figure 6A**) and complete inhibition of icatibant 100 µg/mL-induced wheals at 90 to 200 mg QD (**Figure 6B**)

Figure 4. Skin challenge control measurements of icatibant-induced wheals

Figure 5. Representative skin challenge test wheals of 2 participants pre- and post-therapy with SEP-631 imaged by AllergyScope™

Figure 6. Skin challenge PD results in the Part B (MAD) cohorts at steady-state on Day 9

Conclusions

- SEP-631 was well tolerated across all studied doses, with no clinically significant adverse events
- The PK profile of SEP-631 supports once-daily oral dosing
- No clinically meaningful effect of food on SEP-631 exposure was observed, supporting SEP-631 dosing without food restriction
- Use of the AllergyScope™ detector enabled accurate assessment of skin test wheals that was consistent with, but more precise than, historical measurements, representing a significant advancement over conventional measurement techniques
- SEP-631 treatment resulted in robust inhibition of icatibant-induced skin wheal formation across all studied doses, with complete inhibition observed at doses as low as 10 mg QD (10 µg/mL icatibant challenge)
- Inhibition of skin wheals suggests that SEP-631 engages with the target, MRGPRX2, and inhibits downstream signaling, thereby blocking mast cell degranulation
- These Phase 1 results in healthy adult volunteers support further clinical evaluation of SEP-631 in patients with mast cell-driven diseases, such as CSU

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Disclosures

JRT and **DWM** are co-founders of FLARE Diagnostics. **CCC** has nothing to declare. **MG, LM, JKS, DL, ANL,** and **JBK** are employees of Septerna, Inc., and may hold shares in the company. **HMB** and **EA** were employees of Septerna, Inc., at the time of the study and may hold shares in the company. **SSS** received research grants from Novartis, Allakos, Evomune, Jasper, Celldex, and Blueprint; and received consultancy fees from Allakos, Granular Therapeutics, Genentech, Celldex, Evomune, Novartis, Escient, Celltrion, Sanofi, Nuocor, GSK, and Septerna, Inc.